Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces

Class Test One SOLUTIONS **29 March 2018** Total marks: 40

1. Let $A = [0, 4) \cup (10, 20]$.

   (a) $\text{int}(A) = (0, 4) \cup (10, 20)$

   (b) $\text{bd}(A) = \{0, 4, 10, 20\}$

   (c) $\text{int}(A) = A$

   (d) $\text{bd}(A) = \emptyset$ [1+1+1+1=4]

2. Suppose (X, d) is a metric space. Define ρ by:

$$\rho(x, y) = g(d(x, y)) \text{ for all } x, y \in X$$

   Note that ρ is well-defined, because, for each $x, y \in X$, $d(x, y) \in [0, \infty)$, so $\rho(x, y) = g(d(x, y)) \in [0, \infty)$ also.
   For all $x, y, z \in X$:

   (M1) $\rho(x, x) = g(d(x, x)) = g(0) = 0$. This uses (a) and the property (M1) for d.
   If $\rho(x, y) = 0$, then $g(d(x, y)) = 0$, so $d(x, y) = 0$ and so $x = y$. This uses (a) and the property (M1) for d.

   (M2) $\rho(x, y) = g(d(x, y)) = g(d(y, x)) = \rho(y, x)$. This uses (M2) for d.

   (M3) We know $d(x, z) \leq d(x, y) + d(y, z)$ by the property (M3) for d.
   Using (b) gives $\rho(x, z) = g(d(x, z)) \leq g(d(x, y) + d(y, z))$.
   Using (c) then gives $g(d(x, y) + d(y, z)) \leq g(d(x, y)) + g(d(y, z))$.
   Putting this together gives $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.

   [6]

3. In the metric space $(\mathbb{R}^2, \delta)$, provide a sketch of $B((0, 0); 10)$, that is, the open ball centred at $(0, 0)$ with radius 10.

   The open ball is
   $\{(x, y) \in \mathbb{R}^2 : \delta((0, 0), (x, y)) < 10\}$
   $= \{(x, y) \in \mathbb{R}^2 : \max\{|x|, 5|y| < 10\}$
   $= \{(x, y) \in \mathbb{R}^2 : -10 < x < 10 \text{ and } -2 < y < 2\}$.

   [3]

4. (a) $\text{bd}(A) = \overline{A} \cap \overline{A'}$

   (b) From line (3), for all $n \in \mathbb{N}$, $z_n \in B(x; \frac{1}{n})$, so $d(z_n, x) < \frac{1}{n}$. Since $\frac{1}{n} \to 0$ as $n \to \infty$, $d(z_n, x) \to 0$, so $z_n \to x$.

   (c) From line (4), we have that $x = \lim z_n$. But, from line (3), $z_n \in A'$ for all n. By the definition of closure, $x \in \overline{A'}$.

   (d) From line (1), $x \in A$. From line (5), $x \in \overline{A'}$. So $x \in A \cap \overline{A'} \subseteq \overline{A} \cap \overline{A'} = \text{bd}(A)$.

[8]

5. (a) True.
   Suppose (x_n) is a sequence in X such that $x_n \to x$ in (X, d) and $x_n \in F \cap K$ for all $n \in \mathbb{N}$.
   Then $x_n \to x$ and $x_n \in F$ for all n, so, since F is closed, $x \in F$.
   But also $x_n \to x$ and $x_n \in K$ for all n, so, since K is closed, $x \in K$.
   So $x \in F \cap K$, as required.

   (b) False.
   In $(\mathbb{R}, d_E)$, let $A = \mathbb{Q}$ and $B = \mathbb{R} \setminus \mathbb{Q}$.
   Then $A \cap B = \emptyset$, so $\overline{A \cap B} = \emptyset$.
   But $\overline{A} \cap \overline{B} = \mathbb{R} \cap \mathbb{R} = \mathbb{R}$.

[3+4=7]

6. Let (X, d) be a metric space, $x \in X$ and A a non-empty subset of X.

   (a) $\text{dist}(x, A) = \inf\{d(x, a) \,|a \in A\}$.

   (b) $x \in \overline{A} \iff$ there exists a sequence (a_n) such that $a_n \in A$ for all n and $a_n \to x$ in (X, d).

   (c) Suppose that $\text{dist}(x, A) = 0$.
   Then $\inf\{d(x, a) \,|a \in A\} = 0$, so no strictly positive number can be a lower bound of the set $\{d(x, a) \,|a \in A\}$.
   So, for all $n \in \mathbb{N}$, there exists $a_n \in A$ such that $d(x, a_n) < \frac{1}{n}$.
   Then $a_n \to x$, and so $x \in \overline{A}$. [2+2+4=8]

7. (a) We prove the contrapositive of the desired result.
So suppose that, for all $x \in X$, and all $r > 0$ we have $B(x;r) \neq X$.
We provide a sequence which has no convergent subsequence.

First we note that the union of finitely many open balls is always contained in a larger open ball:
$B(a_1;r_1) \cup B(a_2;r_2) \subseteq B(a_1;r)$ where $r = r_1 + r_2 + d(a_1,a_2)$ because,
if $z \in B(a_1;r_1)$ then $d(z,a_1) < r_1 \leq r$, and
if $z \in B(a_2;r_2)$ then $d(z,a_1) \leq d(z,a_2) + d(a_2,a_1) < r_2 + d(a_1,a_2) \leq r$.
Our assumption therefore also guarantees that no finite union of open balls is all of X.

Form the desired sequence as follows.
For x_1 choose any element of X.
Choose $x_2 \notin B(x_1;1)$. This is possible, by assumption.
Choose $x_3 \notin B(x_1;1) \cup B(x_2;1)$.
Etc: choose $x_n \notin B(x_1;1) \cup B(x_2;1) \cup \ldots \cup B(x_{n-1};1)$.
Note that $d(x_i,x_j) \geq 1$ for all $i \neq j$, so (x_n) has no convergent subsequence.

(b) Yes, it is possible. Here are two possibilities, obviously not the only ones:

i. Let $X = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$, with the metric d_E.
In this space, $\{x\}$ is open, for all $x \neq 0$.
So, for a subset A of X, if $0 \notin A$, then A is open.
If $0 \in A$, then $0 \notin A'$, so A' is open, so A is closed.
However, not every subset is clopen, because $\{0\}$ is not open.

ii. Let $X = \mathbb{C}$ and d be the metric defined by

$$d(z,w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w. \end{cases}$$

In this space, $\{a + bi\}$ is open, for all $a + bi \neq 0$.
So, for a subset A of X, if $0 \notin A$, then A is open.
If $0 \in A$, then $0 \notin A'$, so A' is open, so A is closed.
However, not every subset is clopen, because $\{0\}$ is not open.

[2+2=4]